

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 41

Level 3 — Advanced · Number Sense I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 3 — Advanced	Topic Focus	Number Sense I — Place-Value Reasoning & Digit Puzzles
Questions	20	Time Allowed	40 minutes
Total Marks	20	Worksheet No.	41 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1–5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Welcome to Level 3 — Advanced. Worksheet 41 raises Number Sense to genuine reasoning: 6-digit and lakh-place numbers, palindromes, working BACKWARDS through a chain of operations, double-condition rounding puzzles, consecutive-number sums, and digit-frequency counting. These match the harder Mathematical-Reasoning questions of the SOF IMO.

Learning Objectives

- Reason about place values in 5- and 6-digit numbers (Indian system).
- Solve digit puzzles: palindromes, equal-digit and digit-sum conditions.
- Work backwards through chains of add/subtract/multiply.
- Handle double-condition rounding and 'midway' problems.
- Count digit frequencies and numbers within ranges.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

- Q1.** In **8,47,562** (Indian system), what is the SUM of the digit in the lakhs place and the digit in the thousands place?
- (A) 11 (B) 15
(C) 12 (D) 9
- Q2.** A 5-digit number reads the same forwards and backwards (a palindrome). Its first digit is 7, its middle digit is 0, and its second digit is 3. What is the number?
- (A) 73037 (B) 70307
(C) 73037 reversed (D) 73073
- Q3.** What is the LARGEST 5-digit number with all DIFFERENT digits?
- (A) 99,999 (B) 98,765
(C) 98,760 (D) 97,531
- Q4.** The difference between the greatest and the smallest 4-digit numbers formed using **5, 0, 3, 8** (each used once) is:
- (A) 5,472 (B) 5,427
(C) 5,283 (D) 5,463
- Q5.** How many whole numbers lie between 1,000 and 2,000 that have **5** in the hundreds place?
- (A) 10 (B) 100
(C) 99 (D) 1
- Q6.** When a number is multiplied by 10, then 100 is added, the result is 4,100. The ORIGINAL number is:
- (A) 390 (B) 400
(C) 410 (D) 41
- Q7.** A number rounded to the nearest THOUSAND is 7,000 and rounded to the nearest HUNDRED is 6,500. The number could be:
- (A) 6,549 (B) 6,450
(C) 6,500 only (D) 7,400
- Q8.** The sum of three consecutive numbers is 72. The SMALLEST of them is:
- (A) 22 (B) 23
(C) 24 (D) 25
- Q9.** What is the place value of 9 in the product **9 × 1,000**?
- (A) 9 (B) 900
(C) 9,000 (D) 90



Q10. A 3-digit number has digits that add up to 18. The hundreds digit is 9 and the tens digit is 4. The ones digit is:

- (A) 3 (B) 4
(C) 5 (D) 6

Q11. Which number is exactly midway between 2,400 and 3,000?

- (A) 2,600 (B) 2,700
(C) 2,650 (D) 2,750

Q12. If you count by 7s starting from 3 (i.e. 3, 10, 17, 24 ...), which of these numbers WILL you say?

- (A) 50 (B) 52
(C) 51 (D) 53

Q13. The greatest 6-digit number is:

- (A) 9,99,999 (B) 1,00,000
(C) 99,999 (D) 10,00,000

Q14. A number becomes 5,000 when you ADD 1,250 to it and then SUBTRACT 750. The number is:

- (A) 4,500 (B) 4,250
(C) 5,500 (D) 3,000

Q15. In a number, the ten-thousands digit is twice the thousands digit, the thousands digit is 3, and all other digits are 0. The number is:

- (A) 63,000 (B) 36,000
(C) 60,300 (D) 6,300

Q16. How many times does the digit 2 appear in the numbers from 1 to 30?

- (A) 11 (B) 12
(C) 13 (D) 14

Q17. The number 'forty thousand forty' is written as:

- (A) 40,040 (B) 40,400
(C) 44,000 (D) 4,040

Q18. A number is greater than 4,995 and less than 5,005. How many such WHOLE numbers are there?

- (A) 8 (B) 9
(C) 10 (D) 11

Q19. I am a 4-digit number. My digits are all the same. I am divisible by 3. The SMALLEST such number is:

- (A) 1,111 (B) 2,222
(C) 3,333 (D) 1,113



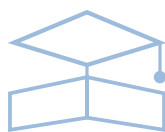
Q20. A number is doubled and then 5 is added, giving 27. The ORIGINAL number is:

(A) 10

(B) 11

(C) 16

(D) 12



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	A	B	A	B	B	A	B	C	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	B	A	A	A	C	A	B	C	B

Step-by-Step Solutions

Q1. Correct answer: **(B)**

8,47,562 → 8 lakhs, 4 ten-thousands, 7 thousands, 5 hundreds, 6 tens, 2 ones. Lakhs digit 8 + thousands digit 7 = **15**.

Q2. Correct answer: **(A)**

A palindrome of form ABCBA: A=7, B=3, C=0 → 7-3-0-3-7 = **73037**.

Q3. Correct answer: **(B)**

Use the biggest digits, all different, in decreasing order: 9, 8, 7, 6, 5 → **98,765**.

Q4. Correct answer: **(A)**

Greatest = 8,530; smallest (cannot start with 0) = 3,058. Difference = 8,530 – 3,058 = **5,472**.

Q5. Correct answer: **(B)**

These are 1,500 to 1,599 — that is **100** numbers.

Q6. Correct answer: **(B)**

Work backwards: 4,100 – 100 = 4,000; then 4,000 ÷ 10 = **400**.

Q7. Correct answer: **(A)**

Round-to-thousand 7,000 → number is 6,500–7,499. Round-to-hundred 6,500 → number is 6,450–6,549. Both true for 6,450–6,549. Only **6,549** fits (6,450 rounds to 6,000 for thousands).

Q8. Correct answer: **(B)**

The middle number = 72 ÷ 3 = 24. So the three are 23, 24, 25 → smallest is **23**.

Q9. Correct answer: **(C)**

9 × 1,000 = 9,000, in which 9 is in the thousands place: place value **9,000**.

Q10. Correct answer: **(C)**

9 + 4 + ones = 18, so ones = 18 – 13 = **5**.

Q11. Correct answer: **(B)**

Midway = (2,400 + 3,000) ÷ 2 = 5,400 ÷ 2 = **2,700**.

Q12. Correct answer: **(B)**

Numbers said leave remainder 3 when divided by 7. 52 ÷ 7 = 7 r 3 ✓. So **52** is in the list.

Q13. Correct answer: **(A)**

The greatest 6-digit number is **9,99,999** (one less than ten lakh).

Q14. Correct answer: **(A)**

Work backwards: 5,000 + 750 = 5,750; then 5,750 – 1,250 = **4,500**.

**Q15. Correct answer: (A)**

Thousands = 3, ten-thousands = $2 \times 3 = 6$. Number = 6 (ten-thousands), 3 (thousands), 0,0,0 → **63,000**.

Q16. Correct answer: (C)

Ones place 2s: 2,12,22 → 3. Tens place 2s: 20–29 → 10. But 22 was counted once in each group, so total = $3 + 10 = 13$.

Q17. Correct answer: (A)

Forty thousand = 40,000; forty = 40. Total = **40,040** (zeros in hundreds and... 40,040).

Q18. Correct answer: (B)

Numbers from 4,996 to 5,004: count = $5,004 - 4,996 + 1 = 9$.

Q19. Correct answer: (C)

Equal-digit numbers: 1111 (digit sum 4, not $\div 3$), 2222 (sum 8, no), **3333** (sum 12, yes $\div 3$). Smallest that works is 3,333.

**Q20. Correct answer: (B)**

Work backwards: $27 - 5 = 22$; $22 \div 2 = 11$.

